

1 Data

The datasets we analyzed were collected in exposure-test experiments on cross-talker and cross-accent generalization during the perception of L2-English speech by L1-English listeners: AN19 [?], B23 [?], and X21 [?]. The datasets are among the largest, publicly available datasets of this type. Each dataset consists of the exposure and test stimuli used in the experiment—necessary to derive predictions from our DNN-based approach—and measures of the human behavior we seek to understand and predict: how well participants recognized the speech presented during test, depending on the speech they experienced during exposure.

During the exposure phase, listeners are randomly assigned to an exposure condition. Exposure conditions manipulate what type of speech listeners experience—e.g., whether listeners hear L1-English or L2-English, whether the L2 English comes from talkers of one or more different L1 language background, and whether listeners hear speech from one or more talkers. Following exposure, the test phase assesses how well listeners recognize different types of L2-English. This test phase is held constant across the different exposure conditions.¹

1.1 AN19: Alexander and Nygaard (2019), Experiment 2

We used Experiment 2 from Alexander and Nygaard [?]. The experiment measured within- and cross-accent generalization across talkers in 8 different between-participant conditions ($N = 160$ participants, ≥ 20 per condition). The design crossed 4 exposure conditions (control, same accent, different accent, mixed accent) with 2 test conditions (L1-Korean L2-English vs. L1-Spanish L2-English; henceforth Korean- and Spanish-accented English). All conditions tested cross-word cross-talker generalization after multi-talker exposure.

1.1.1 Exposure phase

Listeners heard isolated word recordings of 144 different words (24 per talker) and transcribed them. Listeners in the *control* condition received no exposure. Listeners in the *same accent* condition heard 6 talkers (3 female, 3 male) of the same accent as during test (i.e., Korean- or Spanish-accented English). Listeners in the *different accent* condition heard 6 talkers (3 female, 3 male) of the ‘opposite’ accent as during test: listeners who heard Korean-accented English during test heard Spanish-accented English during exposure and vice versa. Listeners in the *mixed accent* condition heard 6 talkers (3 female, 3 male) of six different L1 backgrounds, none of which was L1-Spanish or -Korean.

We additionally obtained recordings of the same 144 words from XXX L1 talkers of English from Alexander and Nygaard. These recordings were not used in the experiment but allow us to calculate the relevant quantities hypothesized by the HVE and SBI to predict generalization in the *control* condition (see 2).

¹While all of the experiments we analyzed have multiple different test conditions—assessing the recognition of different types of L2-English—the distribution of different exposure conditions was balanced within each of these different test conditions. That is, all three data sets we selected to analyze avoid confounding the effects of exposure and test.



Fig. 1: Listeners’ transcription accuracy during test in Alexander and Nygaard (2019), Experiment 2 [?] depending on the exposure condition, reanalyzed using the same GLMM used to evaluate the HVE and SBI (Binomial with log-link).

1.1.2 Test phase

During test, listeners heard isolated word recordings of 48 different words (8 per talker) and transcribed them. Listeners in the *Korean-accented* test condition heard 6 Korean-accented talkers (3 female, 3 male). Listeners in the *Spanish-accented* test condition heard 6 Spanish-accented talkers (3 female, 3 male). Test talkers and words were never the same as those heard during exposure. Generalization was measured as the proportion of correctly transcribed words during test.

1.1.3 Replication of original findings

Alexander and Nygaard [?] found that listeners in the *same accent* condition achieved significantly higher accuracy during test than in the *control* and *different accent* conditions. Additional post-hoc comparisons found that *mixed accent* exposure yielded numerically higher accuracy than *control* or *different accent* exposure. For the Spanish-accented test, *mixed accent* exposure achieved accuracy comparable to that of *same accent* exposure, whereas it achieved lower accuracy than *same accent* exposure for the Korean-accented test, as shown in Figure 1.

1.1.4 Exposure-test combinations

One feature of the experiment makes it particularly suitable for the present purpose. Most experiments on cross-talker and cross-accent generalization employ a single exposure and/or test condition of each type—e.g., a single Korean-accented test condition and a single Spanish-accented test condition. This makes it impossible to distinguish between the perceptual consequences of talker-specific and accent-specific pronunciation features. Alexander and Nygaard instead used two variants of each test condition, i.e., two sets of 6 talkers for each the Korean- and the Spanish-accented test conditions.

Similarly, two variants were created for each exposure condition. For the *same accent* and *different accent* exposure conditions, whichever six talkers were not used during test were used during exposure. For the *mixed accent* exposure condition, two different mixes of six different L2-accented talkers were used. The 12 talkers for this condition came from 12 different L1 backgrounds (Albanian, Bengali, Dutch, French, German, Hindi, Japanese, Mandarin, Romanian, Russian, Somali, and Turkish).

In total, the data thus contain $4 \text{ exposure conditions} \times 2 \text{ test conditions} \times 6 \text{ test talkers each} \times 2 \text{ set of test talkers} = \mathbf{96 \text{ unique exposure-test combinations}}$ that differ in the accents, talkers, and words that listeners heard during exposure and test. This yields $96 \times 8 = \mathbf{768 \text{ unique word-level measurements of recognition accuracy}}$ (each with $N = 10$ observations from 10 different participants).

Our analysis approach compares the HVE and SBI hypothesis in terms of their ability to predict these 768 unique observations. This also allows us to test whether the HVE or SBI can explain variability in listeners' perception beyond that $4 \times 2 = 8$ design conditions of the experiment.

1.2 X21: Xie et al. (2021), Experiment 1a

We used Experiment 1a from Xie et al. [?]. The experiment is a large-scale replication of a seminal study on the perception of Mandarin-accented L2-English by L1-English listeners by Bradlow and Bent [?]. The experiment measured within-accent generalization in 4 different conditions ($N = 320$ participants, 80 per condition). Like for the AN19 data, all conditions assessed cross-word generalization. Unlike the AN19 data, conditions differed in whether they involved exposure to a single or multiple talkers, and all conditions focused on generalization within the same accent.

1.2.1 Exposure phase

During the exposure phase, participants heard and transcribed 80 recordings of 16 different sentences (5 recordings / sentence), taken from the ALLSSTAR database [?].

Recordings differed between exposure conditions: 16 sentences each from five different L1-English talkers (*control* exposure), the same but from five Mandarin-accented talkers different than the test talker (*within-accent multi-talker*), the same but from a single Mandarin-accented talker different from the test talker (*within-accent single-talker*), or the same but from the same Mandarin-accented talker as during test (*talker-specific*).



Fig. 2: Listeners’ transcription accuracy during test in Xie et al. (2021) [?] depending on the exposure condition, reanalyzed using the same GLMM used to evaluate the HVE and SBI (Binomial with log-link).

1.2.2 Test phase

During test, all groups of participants transcribed 16 sentence recordings from a single Mandarin-accented talker, taken from the ALLSTAR database [?]. Transcription accuracy was measured for the 3-4 content words per sentence, yielding 51-52 scored keywords per participant (e.g., “boy”, “fell”, “window” for the sentence “the boy fell from the window”). Sentences were never the same as those heard during exposure. Generalization was measured as the proportion of correctly transcribed words during test.

1.2.3 Replication of original findings

Xie et al. found that accuracy during test was highest in the talker-specific condition. Accuracy in both the multi-talker and single-talker conditions was lower than in the talker-specific condition but higher than in the control condition, as shown in Figure 2.

1.2.4 Exposure-test combinations

Similar to the AN19 data, the X21 data are particularly suitable for the present purpose because they contained a large number of exposure-test combinations.

Specifically, Xie et al repeated the design described above for four different Mandarin-accented test talkers. Additionally, the design counterbalanced which of two sets of 16 sentences were used during during test. This resulted in **64 unique exposure-test combinations**: four combinations of exposure- and test-talker for the control, multi-talker, and talker-specific conditions, and 20 variants of the single-talker conditions, each repeated equally often for each of the two sets of test sentences. With 51-52 keywords per test, the data contain a total of **3296 unique word-level measurements of recognition accuracy**.²

1.3 B23: Bradlow et al. (2023)

Bradlow et al. [?] combined the design of experiments on within-accent cross-talker generalization [?] with the design of experiments on cross-accent cross-talker generalization [?]. The experiment contained a total of 9 different exposure conditions that measured generalization within and across L2-English accents after exposure to one or more talkers ($N = 195$ participants, ≥ 20 per condition).

The design crossed the exposure conditions with 4 within-participant test conditions, each consisting of one talker from a different variety of L2 English (Brazilian Portuguese-, Farsi-, Spanish-, or Turkish-accented English). Like English, all four of the L1 languages are part of the Indo-European language family. Within that language family, however, the four languages were genealogically and areally heterogeneous, and differed substantially in their segmental and supra-segmental phonologies, as well as many other linguistic properties (for details [?]). As in the AN19 and X21 data, all conditions assessed cross-word generalization. All stimuli were taken from the ALLSTAR database [?] and were of similar complexity as those used in the X21 data.

1.3.1 Exposure phase

In the *control* condition, participants did not receive any exposure and just transcribed the test sentences. For all other conditions, participants transcribed 60 different exposure sentences prior to the test phase. Four different *single-talker* conditions each employed recordings from only one of the four test talkers that differed in their L1 background (Brazilian Portuguese-, Farsi-, Spanish-, or Turkish-accented English). Four different *multi-talker* conditions each employed recordings from all but one of the four test talkers (20 sentence recordings each).

1.3.2 Test phase

During test, all groups of participants transcribed 60 sentence recordings, 15 each from four talkers of the four different L2-English accents. For the *single-talker* exposure conditions, the test data from the same talker thus measured *talker-specific* adaptation, whereas the test data from the three other test talkers measured *cross-accent single-talker* generalization. For these *multi-talker* exposure conditions, all but one of the four test talkers thus measured *talker-specific* adaptation in the presence of additional

²The X21 data were not balanced at this level, so that some word-level measurement have observation from more participants than others. The GLMM analyses we conduct are not affected by this, as their random effect structure adequately captures these imbalances.



Fig. 3: Listeners’ transcription accuracy during test in Bradlow et al. (2023), Experiment 1a [?] depending on the exposure and test condition, reanalyzed using the same GLMM used to evaluate the HVE and SBI (Binomial with log-link).

exposure, while the fourth test talker measured *cross-accent multi-talker* generalization. As in the X21 data, generalization was measured as the proportion of correctly transcribed words during test. However, only sentence-level accuracies—the number of correctly transcribed keywords for each sentence—were stored.

1.3.3 Replication of original findings

Compared to the other two data sets, the results reported in [?]—replicated in Figure 3—are more heterogeneous. Compared to the control group, neither multi- nor single-talker exposure reliably improved accuracy during test. Even talker-specific exposure did not reliably outperform single- or multi-talker exposure, unlike in almost all previously published work. In some cases, talker-specific exposure barely improved accuracy during test even when compared to the control group. This suggests that the study had low power, as confirmed by auxiliary analyses reported in 3.

1.3.4 Exposure-test combinations

In total, the B23 data yield 9 exposure conditions $\times$ 4 test conditions = **36 unique exposure-test combinations**. With 15 test sentence for each of these

exposure-test combinations, the B23 contains a total of **540 unique sentence-level measurements of recognition accuracy**.

2 Predicting listeners' perception

3 Results for Bradlow et al. (2023)

As reported in the main text, one of the three data sets [?] had very low self-replicability: predicting the responses of held-out listeners from all other listeners in the same condition yielded little signal ($z \leq XXX$). We thus did not report the results for these data in the main text. Even for these data, however, noise-corrected support for the SBI was of similar magnitude as for the other two data sets, though it barely reached significance.

XXX

References

4 Equations

Equations in L^AT_EX can either be inline or on-a-line by itself (“display equations”). For inline equations use the `$...$` commands. E.g.: The equation $H\psi = E\psi$ is written via the command `$H \psi = E \psi$`.

For display equations (with auto generated equation numbers) one can use the `equation` or `align` environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (1)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (2)$$

Notice the use of `\nonumber` in the `align` environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an `align` environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\text{GeV}}\right)^{-3} \left[1 + \frac{3\ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \quad (3)$$

The class file also supports the use of `\mathbb{}`, `\mathscr{}` and `\mathcal{}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsubsection ??).